



## Diversity in Product Design

2022-05-25

### Student Materials

- Handout 1
- Handout 2
- Handout 3

### Overview

Ethics background suggested: Students should be familiar with the four ethical frameworks presented in the first year curriculum (virtue ethics, deontology, consequentialism, analogies). If they (or the professor) are not, brief summaries are provided in the activity section.

Subject matter referred to in this lab: product design

Placement in overall ethics curriculum:

- We recommend that this lab be used after the first year curriculum and in a computer architecture or systems course; this lab could be adapted to an engineering course that considers product design.
- Recommended previous labs: Foundational Labs
- Recommended follow-up labs: No direct follow-up labs.

Time required:

- Out of class: None; although this lab can be adapted to online discussion.
- In class: 40 minutes

Learning objectives:

- Students can recognize the value of a diverse team during product design.
- Students can determine critical members of a design team for a particular product.
- Students reflect on lack of diversity as an ethical issue.
- Students are introduced to methods for encouraging and welcoming diverse voices.

Ethical issues to be considered: Diversity. Engineers have often been rewarded for getting their product to market quickly. While this can be an important goal, often this approach has resulted in products that have limited appeal. One of the reasons for the limited appeal is that a diverse team was not employed in the design. This lab discusses the need for, and



## integrated ethics

ways to build an effective, diverse design and development team. It also considers that lack of diversity can result in an unethical product.

### Flow

- As a class, students are introduced to the idea that a diverse product design team makes a better product.
- In groups, students are given examples of products and asked who should be on the design team for that product and why; consider the ethicality of design decisions and are asked to answer questions regarding their experiences with suggesting unique solutions or with receiving them.
- As a class, reflect on what was discovered and discuss ideas for ensuring as to how to encourage and welcome diverse voices in a design team.

Preparation: Read the entire lab. Make slides or other preferred methods to display information of class reading and questions for the introduction and reflection. Print or otherwise make available (online) the readings from the student handout. It is recommended that you assign one of the 6 readings to a small group of 2-4 students, so plan group divisions in advance.

## Guide for Instructors

### Lesson plan

Introduction (10 minutes)

Read or post for the students:

Several years ago the New York Times published an article about a Google Engineer who was finding it challenging to get around in the NYC subway since he had become wheelchair-bound after an accident. Prior to the accident he loved to tell his friends how cool the subway was, and that it could be used anytime, to get anywhere, by anyone. After the accident he realized that only 1/5 of the subway stops were accessible (had elevators) and frequently those elevators did not work. There were times when he was able to get to the stop that he wanted, but was not able to get back to the sidewalk above due to elevator failure. <https://www.nytimes.com/2017/03/29/opinion/new-york-has-a-great-subway-if-youre-not-in-a-wheelchair.html> Accessed 1/2/21

Prior to the accident incurred by the Google engineer, he thought that the subway was amazing. After the accident he had a different view.



Pose each question in turn and let students answer — some sample answers are provided in italics.

1. What do you think that the subway designers might have included to help wheel-chair bound people?
  - At the very least, a system that shows up front which elevators are out of service.\*
2. Why didn't they think of that?
  - Chances are that they did not have a disabled person on the design team.
3. Would there be an economic benefit to having thought of this sooner?
  - Chances are that it would be easier to have included that system in the original design than to go back and fix it later.
4. Would including more accessibility features have been the "right" thing to do from the beginning?
5. Considering some of the ethical frameworks:
  - a. Virtue Ethics: Would it have been what the virtuous person would have done? Why or why not?

Yes - Would have demonstrated compassion and inclusion.
  - b. Consequentialism: Would it have been the best for the most people? Why or why not?

Yes – would have allowed more people to easily use the subway.

No – would have increased cost for subway development making it less economically accessible.
6. Here are some other examples of products that would have benefitted from diverse design teams:
  - a. Some of the first airbags to be installed in cars failed to protect women because they were built to men's specifications, tested with male crash test dummies, and didn't take the female anatomy into account.
  - b. The first voice recognition programs didn't recognize female voices or many accents because they were built and tested by men and native English speakers.
  - c. You may remember Google Photos' image recognition software labeled two black people as "gorillas."

<https://open.nytimes.com/why-having-a-diverse-team-will-make-your-products-better-c73e7518f677> (Accessed 2021-01-05)



## integrated ethics

Activity (15 minutes)

Divide students into groups of 2-4 where each group is assigned a reading from one of the student handouts. Each reading has a set of discussion questions that the group should discuss. Answering the questions will help prepare them for the class reflection.

### Reflection as a class (15 minutes)

#### Part 1 (7 minutes)

Discuss the following two questions, allowing students to respond based on their reading or experience:

1. Having diverse teams helps to improve a product and increase sales – it's just good business sense. Can you think of any examples from your reading or experience that support this statement?

Here are some ideas for the professor if students aren't coming up with many

- Having a variety of genders and ages on a product design team for tools might reveal that a range of tool weights would be required for comfortable use that appeals to a wider audience – increasing the sales.
  - Having a range of ethnicities on a team that designs dolls for children might result in dolls with skin colors, clothing, and accessories that appeal to a wider audience and increase sales.
  - Having a design team for a web page that includes people who are vision impaired (blind, legally blind, color blind) would make the page more useful to many.
2. A design team lacking in diversity often results in a product that excludes some demographic from its use, is offensive to a particular demographic, or unfair to a particular demographic. What did you find in your reading, and would you consider this to be an ethical issue? Why or why not?
    - Teaching evaluation. Probably did not have teachers or parents on the design team; this would be unfair to teachers.
      - Virtues of fairness, transparency, and others violated.
      - Consequentialism: not best for most as some great teachers are eliminated. Could also argue that it is better for most as it removes some bad teachers who are “nice” people but poor teachers.
    - Cultural issues scenarios: probably did not have native Chinese (in one case) or Saudis (in the other case) on the team.
      - Violates virtues of inclusion, sensitivity, professionalism
    - \*Smartphone blog: Did not have elderly adults or people with processing issues on the design team. More is not always better — it can be confusing!



Violates virtues of inclusion, compassion.

Part 2 (8 minutes)

Ask students to reflect on one of these 2 questions depending on the group with which they most relate and then share answers with the class.

1. If you would consider yourself to be a minority in the tech world, have you ever felt like you were afraid to voice your opinion? Why?
2. If you would consider yourself to be a part of the majority in the tech world, how might you encourage diverse voices to speak up?
  - a. Ensure you have a diverse team to design, build and test your products.
  - b. Allow for channels of anonymous input.
  - c. To avoid making assumptions, accept and pay attention to user feedback.
  - d. Encourage mentor-mentee relationships with diverse partnerships. It might be easier for the more mainstream partner to present the unique idea of the other partner.

Assessment

1. Give an example where a diverse produce design team results in a better product.
  - Any of the scenarios here or others could be described.
2. Name 3 distinct kinds of diversity that are helpful on a design team.
  - Some might include gender, race, religion, age, educational level
3. How can the lack of a diverse design team result in an unethical product? Give an example.
  - See answers in the reflection part of this lab.
4. What are some reasons that diverse voices may remain silent? What are ways to encourage diverse voices?
  - See answers in the reflection part of this lab.
5. The priority for most engineers is to ship their products fast. What might be the danger or that as the only priority?
  - Unethical products or products that limit their audience.

For additional reading

Here is an interesting article on what different cultures want in a phone and phone etiquette for that culture:

- <https://www.cnn.com/2012/09/27/tech/mobile-culture-usage/index.html> Accessed 2021-01-07.

According to Antionette Carroll, in Diversity & Inclusion in Design: Why Do They Matter?



integrated  
ethics  
LABS

Diversity in design means diversity of experience, perspective and creativity—otherwise known as diversity of thought—and these can be shaped by multiple factors including race, ethnicity, gender, age, sexual identity, ability/disability and location, among others.”

\* <https://www.aiga.org/diversity-and-inclusion-in-design-why-do-they-matter> Accessed 2021-01-05.

## Bibliography